Ch 14: Introduction to Database Management Systems (DBMS) and SQLite

Application Program Development

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Fall 2026



Database Management Systems

DBMS

A database management system (DBMS) is software that manages large collections of data.

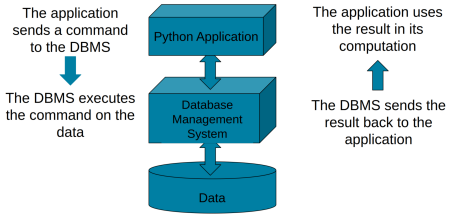


Figure 1: A Python application interacting with a DBMS, which manipulates data (Gaddis, 2021, pg.740)

- Storing data in traditional files has its limits
  - well suited for applications that store only a small amount of data
  - not practical for applications that must store a large amount of data
  - simple operations become cumbersome and inefficient as data increases
- A **database management system** (DBMS) is software that is designed to work with large amounts of data in an efficient and organized manner
  - Data is stored using the DBMS
  - Applications written in Python (or other languages) communicate with the DBMS rather than manipulate data directly
  - DBMS carries out instructions and sends the results back to the application

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SQL and SQLite

SQL stands for **structured query language**

- A standard language for working with database management systems
- Not used as a general programming language
- Consists of several key words, used to construct statements
  - SQL statements are strings passed from the application to the DBMS using API method calls
  - Serve as instructions for the DBMS to carry out operations on its data

In our class assignments we use the `SQLite` database management system

- It is easy to use
- It is automatically installed with Python in a module named `sqlite3`
- Use this import statement

```
import sqlite3
```



Tables, Rows and Columns

Database Organization

Data that is stored in a database is organized into tables. Tables are composed of rows of items, with columns of individual pieces of data.

This row contains data about one person:
Name: Kevin Brown
Phone: 555-9012

Name	Phone
Katie Allen	555-1234
Jill Ammons	555-5678
Kevin Brown	555-9012
Elisa Garcia	555-3456
Jeff Jenkins	555-7890
Leo Killian	555-1122
Marcia Potemkin	555-3344
Kelsey Rose	555-5566

Figure 2: Names and phone numbers stored in a table (Gaddis, 2021, pg.742)

- A DBMS stores data in a **database**
- A database is organized into one or more **tables**
- Each table holds a collection of related data, organized into **rows** and **columns**
  - A **row** is a complete set of data about a single item, divided into columns
  - Each **column** is an individual piece of data about the item

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Column Data Types (1 of 2)

- When you create a database table, you must specify a data type for each column.
  - These are not Python data types, they are specific to the DBMS

SQLite Data Type	Description	Corresponding Python Type
`NULL`	Unknown value	`None`
`INTEGER`	Integer number	`int`
`REAL`	Real number	`float`
`TEXT`	String	`str`
`BLOB`	Binary Large Object	Can be any object



Column Data Types (2 of 2)

Here is a brief summary of each of these data types:

- `NULL` - The NULL data type can be used when the value is unknown or missing. Python reads a `NULL` column value into memory, the value is converted to `None`.
- `INTEGER` - The `INTEGER` data type holds a signed integer value. When Python reads an `INTEGER` column value into memory, the value is converted to an `int` value.
- `REAL` - The `REAL` data type holds a real number. When Python reads a `REAL` column value into memory, the value is converted to a `float` value.
- `BLOB` - The `BLOB` data types holds an object of any type. For example, it can hold an array, or an image. When Python reads a `BLOB` column value into memory, the value is converted to a `bytes` object, which is an immutable sequence of bytes.



Primary Keys

Primary Key

Database tables usually have a **primary key**, which is a column that can be used to identify a specific row.

- A **primary key** is a column that holds a **unique** value for each row in in a database table
- The primary key is used to identify a specific row
- Example employee table:
  - A table stores employee data, and one of the columns holds employee ID numbers, which uniquely identify a single employee in the database
  - Because each employee's ID number is unique, this column can be used as the primary key

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Identity Columns

Sometimes the data you want to store in a table does not contain any unique attributes that can be used as a primary key.

- An **identity column** is a column that contains unique values that are generated by the DBMS
- Identity columns typically contain integers that are autoincremented
- When a table does not have a column that contains unique values, an identity column can be created and used as the primary key

Note: when adding a row that has an identity column in SQLite, you can explicitly provide a value for the identity column. If you provide a value that is already used the DBMS will throw an exception.

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Allowing Null Values

- If a column contains no data, it is said to be null
- Sometimes this is OK
- However a primary key can never be null
- You can apply a constraint to a column to prevent it from being null



Database Connections

Database Connection

Before you can work with a database, you must connect to the database. When you are finished working with the database, you must close the connection.

Typical process of using an SQLite database:

```
Connect to the database
Get a cursor for database
Perform operations on database
Commit changes to database
Close connection to database
```

```
1  import sqlite3
2
3  def main():
4      conn = sqlite3.connect('contacts.db')
5      cur = conn.cursor()
6
7      # Insert code here to perform operations on the
database.
8
9      conn.commit()
10     conn.close()
11
12 # Execute the main function.
14 if __name__ == '__main__':
15     main()
```

The `connect` method returns a `Connection` object for the database.

If the database does not exist, it will be created.

The `commit` method saves changes to the database.

The `close` method closes the connection to the database.

Figure 3: Typical steps to connect and disconnect a database (Gaddis, 2021, pg.742)

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Passing SQL Statements to the DBMS

To execute an SQL statement on an SQLite database:

- Construct a string that holds the SQL statement
- Pass the string to the `Cursor` object's `execute()` method

```
cur.execute(SQLstring)
```

Summary

- A database management system or DBMS is software that manages large collections of data in an efficient and organized manner
  - Applications communicate with the DBMS rather than manipulate data directly
  - DBMS carries out instructions and sends the results back to the application
- SQL or **structured query language** is a standard language for executing commands to manipulate data and communicate with a DBMS
  - SQLite is an example of a light weight implementation of a DBMS that uses SQL
- Data that is stored in a database is organized into tables
  - A row is a complete set of data about a single item
  - Each column is an individual piece of data about the item
- Columns must have a specified DBMS data type associated with it
- A **primary key** is a column that holds a **unique** value for each row in a database table
  - It is used to identify a specific row in a table
- To work with a database, you first connect to it, create a cursor to execute commands with, commit changes, then close the connection when done



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References

Gaddis, T. (2021). *Starting out with Python* (5th). Pearson.

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